

- **Highly customisable:** Without the limitation of OEM widget sets, highly customisable and brand driven UIs can be developed.

The language for development

Dart is the language used for the development of Flutter apps. Even though it has been in use for a long time to create high-quality, critical apps for Android, iOS and the Web, Dart has gained popularity with the introduction of the Flutter framework. Dart is basically a client-optimised language with rich and powerful frameworks along with flexible tooling. It is well suited to reactive programming and supports asynchronous programming through language features and powerful APIs.

If you have never used Dart, fret not! It is a simple language that can be understood quickly and comes with solid documentation. You can start coding in Dart in no time. A new version of it — Dart 2, has been introduced recently with many new features. More about Dart can be found at the website <https://www.dartlang.org/>.

Flutter architecture

Flutter is based on the C++ engine that comprises Skia. The framework is based on this engine and comprises widgets, animation, rendering, etc (source: <https://flutter.io/technical-overview/>).

Setting up the editor

To create your first Flutter app you'll need to set up your editor. For this we need to install the following.

- **Flutter SDK:** This includes Flutter's engine, widgets, tools and the Dart SDK.
- **IDE:** Flutter supports various IDEs like Android Studio, IntelliJ, VS Code, etc. You can work from the command line too.
- **Plugins:** Flutter and Dart plugins must be installed for your particular IDE.

See *Flutter Installation and Setup* on the website for information on how to set up your environment.



Figure 1: Flutter architecture

Problems with current mobile app development methods

Let's start with the problems developers and companies face with the current mobile app development methods.

A majority of mobile application based companies face the problem of maintaining separate teams for Android and iOS. This leads to code duplication, as well as a waste of resources and time for achieving the same goal. With the advent of Flutter, a single team can design applications for both Android and iOS, with unprecedented speed and performance.

The various designer-developer tangles are not new to those in the mobile app development industry. Different platforms have different UIs and it becomes a pain for the developer to not only concentrate on the differing UIs of the platforms but also code the main logic of the application. Simple UI changes like a different colour, font or style can hinder the actual app development as there is a long waiting time for rendering.

Solutions offered by Flutter

Now let's focus on the solutions offered by the amazing Flutter team to these problems.

Flutter offers the same team the capability to design apps for both Android and iOS, with higher speed and better performance, reducing the need for companies to hire extra

manpower. Higher synchronisation can be expected with a single team concentrated on a common goal.

Also, an incredible uniformity can be achieved by the companies when developing their apps if a single team has worked on multiple platforms.

Thus, Flutter can rightfully boast of requiring almost half the development time and half the manpower to achieve twice the efficiency in developing mobile apps.

The Hot Reload feature in Flutter sorts out various designer-developer tangles. By running changes in milliseconds, the UI can be designed in an incredibly fast way, saving the time of the designers and the developers. Changing the colour of a page or an icon can seem too trivial for developers, yet is of utmost important for designers. With Flutter, both teams' goals are achieved by Hot Reload, which basically delivers an updated design in record time.

The only disadvantage of Flutter is that you can now forget about the long coffee break you got while the app was kept to run!

Some big players that use Flutter in production

Google has been using Flutter internally for many of its projects, way before it was opened to the public. Flutter has been used to build Google's mobile sales tool app and